QQ-A-250/17E
March 1, 1971
SUPERSEDING
Fed. Spec. QQ-A-250/17D
March 17, 1967

FEDERAL SPECIFICATION

ALUMINUM ALLOY 7079, PLATE AND SHEET

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

(This specification forms a part of the latest issue of Federal Specification QQ-A-250.)

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the specific requirements for aluminum alloy 7079 plate and sheet; the general requirements are covered in QQ-A-250.

1.2 Classification.

1.2.1 Tempers. The plate and sheet shall be classified as O, T6, T62, T651, or F temper, as specified (see 6.2). The definitions of these tempers shall be as specified in American National Standard ANSI H35.1.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specification:

QQ-A-250 - Aluminum and Aluminum Alloy Plate and Sheet; General Specification for.

Federal Standard:

Fed. Std. No. 184 - Identification Marking of Aluminum, Magnesium, and Titanium.

FSC 9535

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Military Specification:

MIL-I-8950 - Inspection, Ultrasonic, Wrought Metals, Process for.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

American National Standards Institute:

ANSI H35.1 - Alloy and Temper Designation Systems for Aluminum.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, New York 10018.)

3. REQUIREMENTS

3.1 Chemical composition.

3.1.1 The chemical composition shall conform to the requirements specified in table I.

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TABLE I. Chemical composition 1/

Element	Percent	
	Minimum	Maximum
Zinc	3.8	4.8
Magnesium	2.9	3.7
Copper	0.40	0.8
Chromium	.10	.25
Manganese	.10	.30
Iron	-	.40
Silicon	-	.30
Titanium	-	.10
Others, each	-	.05
Others, total	-	.15
Aluminum	Remainder	

1/ Analysis shall regularly be made only for the elements specifically mentioned in the above table. If, however, the presence of other elements is indicated in the course of routine analysis, further analysis shall be made to determine conformance to the limits specified for other elements.

3.2 Mechanical properties.

3.2.1 Mechanical properties of material as supplied. The long transverse mechanical properties shall conform to the requirements of table II. Material shall be capable of meeting the longitudinal and short transverse mechanical properties specified in table III.

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TABLE II. Long transverse mechanical properties

Temper	Thickness	Tensile strength, minimum	Yield strength at 0.2 percent off-set or at extension indicated		Elongation in 2 in. or 4 times D 1/, 2/, minimum
			Minimum	Extension under load	
	Inches	P.s.i.	P.s.i.	Inch/Inch	Percent
O	0.040 thru 0.499	3/ 40,000	3/ 21,000	0.0040	10
	.500 thru 4.000	3/ 40,000	--	--	10
T6	0.015 thru 0.039	73,000	64,000	--	7
	.040 thru .249	74,000	65,000	0.0080	8
T62 4/	0.040 thru 0.249	72,000	62,000	--	8
	.250 thru 1.000	73,000	63,000	--	8
	1.001 thru 1.500	73,000	63,000	--	8
	1.501 thru 2.000	73,000	63,000	--	7
	2.001 thru 2.500	73,000	63,000	--	6
	2.501 thru 3.000	71,000	62,000	--	6
	3.001 thru 4.000	70,000	60,000	--	5
T651	0.250 thru 1.000	74,000	65,000	0.0081	8
	1.001 thru 1.500	73,000	64,000	.0081	8
	1.501 thru 2.000	73,000	64,000	.0081	7
	2.001 thru 2.500	73,000	63,000	.0081	6
	2.501 thru 3.000	72,000	62,000	.0080	6
	3.001 thru 4.000	70,000	60,000	.0078	5
	4.001 thru 4.500	68,000	58,000	.0076	5
	4.501 thru 5.000	68,000	58,000	.0076	5
	5.001 thru 5.500	67,000	57,000	.0075	4
	5.501 thru 6.000	66,000	56,000	.0074	4
F	All	5/	5/	5/	5/

1/ Not required for material 1/2 inch or less in width.
2/ D represents specimen diameter.
3/ Maximum.
4/ Material in the T62 temper is not available from the materials producers.
5/ No requirement.

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TABLE III. Mechanical properties capabilities

Temper	Thickness	Direction of Test	Tensile strength, minimum	Yield strength at 0.2 percent offset, minimum	Elongation in 2 in. or 4 times D, minimum 1/
	Inches		P.s.i.	P.s.i.	Percent
T651	3.001 thru 4.000	Longitudinal	70,000	60,000	6
		Short transverse	65,000	56,000	2
	4.001 thru 4.500	Longitudinal	68,000	58,000	6
		Short transverse	63,000	54,000	2
	4.501 thru 5.000	Longitudinal	68,000	58,000	5
		Short transverse	63,000	54,000	2
	5.001 thru 5.500	Longitudinal	67,000	57,000	4
		Short transverse	62,000	53,000	2
	5.501 thru 6.000	Longitudinal	66,000	56,000	4
		Short transverse	61,000	52,000	2

1/ D represents specimen diameter.

3.2.2 Mechanical properties after heat treatment. In addition to conforming to the requirements of 3.2.1, material in the annealed (O) and the as-fabricated (F) tempers shall, after proper solution heat treatment and aging, also conform to the requirements of table II for the T62 temper. Material as received in the T6 and T651 tempers shall, after proper re-solution heat treatment and aging, be capable of conforming to the requirements specified in table II for the T62 temper.

3.2.3 Bend test. Bend-test specimens taken from material shall be capable of withstanding, without cracking, the bend test specified in QQ-A-250. The values for bend factor N are given in table IV.

TABLE IV. Bend test factor "N"

Thickness	Tempers	
	O	T6 and T62
Inch		
0.015 thru 0.020	1	7
.021 thru .063	2	8
.064 thru .091	3	9
.092 thru .125	4	10
.126 thru .249	5	11
.250 thru .499	6	14

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3.3 Internal defects. When specified (see 6.2), plate shall be ultrasonically inspected (see QQ-A-250). Acceptance limits shall be as specified in table V.

TABLE V. Ultrasonic discontinuity acceptance limits 1/

Thickness	Maximum weight per piece	Discontinuity class 2/
Inches	Pounds	
0.500 thru 1.499	2,000	B
1.500 thru 3.000	2,000	A
3.001 thru 4.500	2,000	B

1/ Discontinuities in excess of those listed in table V may be allowed subject to the approval of the procuring activity, if it is established that they will be removed by machining or that they are in noncritical areas.
2/ The discontinuity class limits are defined in MIL-I-8950.

3.4 Marking. In addition to the marking required in Fed. Std. No. 184, material in the T6 and T651 tempers shall be identified by a lot number marked in at least one location on each piece.

4. QUALITY ASSURANCE PROVISIONS
(See QQ-A-250.)

4.1 Number of tests after heat treatment. From material in the annealed (O) and as-fabricated (F) tempers, an additional number of specimens equal to that required by QQ-A-250 shall be taken and tested after solution heat treatment and artificial aging, to determine compliance with 3.2.2.

5. PREPARATION FOR DELIVERY
(See QQ-A-250.)

6. NOTES

6.1 Intended use. This alloy is intended for use where high strength is required in large section size. This alloy is noted for its high transverse properties, especially its high elongation values. The corrosion resistance of this alloy is comparable to other high strength aluminum alloys.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.

(b) Form and temper required (see 1.2.1).

(c) Dimensions required.

(d) Requirements for sizes not specifically covered (see QQ-A-250).

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(e) Whether ultrasonic inspection of plate is required (see 3.3).

(f) Selection of applicable levels of perservation, packaging, and packing required, if other than level C (see QQ-A-250).

6.3 Temper. Temper T651 is available in plate only.

MILITARY CUSTODIANS:
Army - MR
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INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

QQ-A-250/17E ALUMINUM ALLOY 7079, PLATE AND SHEET

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

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